Smart Forest Fire Detection System

Higher National Diploma in Software Engineering

Final Project Dissertation

2025.1F

School of Computing and Engineering

National Institute of Business Management

Colombo 07

SMART FOREST FIRE DETECTION SYSTEM

Group 04

Group Members:

COHNDSE251F-027 P.D.D.S. Kulasekara

COHNDSE251F-029 H.H.A.M. Sethumika

COHNDSE251F-030 C. H. Amarasinghe

COHNDSE251F-031 W. R. D. I. Sandeepani

Higher National Diploma in Software Engineering
2025.1F
School of Computing and Engineering
National Institute of Business Management
February 2026

Declaration

The project is submitted in partial fulfillment of the requirement of the Higher National Diploma in Software Engineering (Batch 2025.1F) of the National Institute of Business Management.

Detailed Declaration:

"We hereby declare that this dissertation represents our own work, conducted under the supervision of Mr. Sampath Gamage, and has not been previously included in a thesis or dissertation submitted to this or any other institution for a degree, diploma, or other qualifications. We have adhered to the ethical guidelines of National Institute of Business Management (NIBM) and acknowledge all contributions made towards the completion of this research."

Member Details:

Index	Name	Signature
COHNDSE251F-027	P. D. D. S. Kulasekara	
COHNDSE251F-029	H.H.A.M Sethumika	
COHNDSE251F-030	C. H. Amarasinghe	
COHNDSE251F-031	W.R.D.I. Sandeepani	

Table of Content

List of Keywords and Figures.

List of Keywords

Keywords:
Internet of Things (IoT), Smart Forest Guardian, Forest Fire Detection, Autonomous Robot, ESP32, Severity Level Classification, Machine Learning, Cloud Computing, GPS Tracking, Smart Alert System.

List of Abbreviations

Abbreviation	Description
IoT	Internet of Things
ESP32	Wi-Fi Enabled Microcontroller
GPS	Global Positioning System
ML	Machine Learning
MQ-9	Gas Sensor Module
API	Application Programming Interface
UI	User Interface

Tables

Figures

Chapter 1: Introduction

1.1 Domain Introduction

Controlling forest fires is a very essential aspect of disaster management and environmental surveillance in the world. Forest fires are devastating to the ecosystems, wildlife and human life as well as national economies. As the impacts of global warming continue to intensify and dry periods become longer and longer, the number and severity of forest fires is becoming alarming all over the world.

Traditional methods of fire detection, such as manual patrols, satellite surveillance and stationary camera systems are susceptible to slow detection and lack of ground level coverage. These technologies are expensive to sustain, have no predictive qualities and especially in isolated and thickly wooded zones where fire can propagate at an early stage without being detected until later when it may have extended to vast proportions.

The project that is meant to address these shortcomings is the Smart Forest Guardian. It is a self-driven, IoT-based, and solar-powered robotic system that is meant to constantly survey forest environments against the risk of fire. The system, utilizing IoT sensors, robotics, cloud computing, machine learning, and renewable solar energy, is a long-term, smart, and reliable solution to the problem of wildfire prevention and management in the world.

1.2 Research Gap

An analysis of the current literature shows that there are a number of shortcomings of the current technologies used to detect forest fires. According to Kumar (2020), one sensor-based fire detection system proposed by the author involves the use of temperature sensors and flame sensors, and this sensor could not anticipate any risk, as it would detect fire only after ignition. Liu (2019) also designed a forest monitoring robot that had GPS tracking capabilities, but there were no severity classification systems, as well as sustainable sources of power, which restricted the ability to use it in remote locations in the long term.

Rao (2022) discussed the use of gas sensors in the monitoring of wildfires and was only interested in sensor accuracy, without discussing autonomous movement and real-time notification. Jain and Sharma (2021) suggested machine learning models to predict wildfire occurrence by satellite data but the model used aerial data to predict wildfires and as such, it was less granular and delayed to respond.

Moreover, the vast majority of the available systems rely on traditional sources of power or dysfunctional battery, which makes them less viable in the long run in the off-grid forest settings. The identified research gap is the lack of a single platform that integrates all the following features into a single autonomous system: solar-powered operation, multi-sensor ground-level detection, intelligent prediction of fire risks, categorizing their severity, and cloud-based real-time warning.

1.3 Research Problem Definition.

The existing forest fire surveillance systems do not offer early fire detection, severity-based risk analysis and autonomous ground level monitoring as well as sustainable long-term power solutions that are applicable in remote forest locations. It leads to slow reaction to the emergency and high ecological and economic losses.

1.4 Research Objectives.

The following objectives are the objectives of this research:

- Designing and developing a self-driven IoT-based robotic system operated using solar energy to monitor forest fires continuously.
- To install a multi-sensor fire detection system based on temperature, humidity, gas concentration and flame sensor.

- To design a Severity Level Classification System that should classify fire risk into Low, Medium, and High.
- To incorporate machine learning algorithms to forecast fire risks in real-time using sensor data of past and present.
- To offer real-time location tracking using GPS and cloud-based notification using a web dashboard and mobile app.
- To minimize the false alarm by sensor fusion approach and threshold-based decision making.
- To gain this sustained and unbroken operation of a system using renewable solar energy in off-grid forests set-ups.

1.5 Proposed Solution.

The solution that is proposed is an independent robotic platform that is solar powered, has multiple environmental sensors on it, a GPS module and wireless communication capabilities. The system is to patrol specific forest locations and relay real-time data on the environment to a centralized cloud infrastructure.

The ESP32 microcontroller is used as the central processing unit, and it will be involved in acquiring sensor data and sending it to the cloud, via Wi-Fi. Some of the parameters that are checked on the environment are temperature, humidity, gas concentration, flame presence, light intensity and rainfall. The deployed machine learning model uses analysis of real-time and historical sensor data to forecast the risk level of fire.

The Severity Level Classification System assigns a risk as Low, Medium, or High depending on pre-established sensor limits and ML model predictions, and allows authorities to prioritize and attract their emergency responses effectively. Real-time surveillance, GPS tracking of fire location, alerts, and historical data analysis are offered on the basis of a web-based dashboard and mobile application. Solar panel and rechargeable battery system make it possible to operate continuously without any maintenance even in most isolated forest areas.

1.6 Chapter Summery.

In this chapter, the project domain was presented, the essential gaps in the current literature of forest fire detection were identified, the research problem was outlined, the objectives of the research were stated, and the suggested solution was a Smart Forest Guardian. The system is the first in terms of its combination of renewable solar energy, autonomous robotics, multi-sensor fusion, severity classification, and predictive analytics to provide a comprehensive, sustainable, and intelligent wildfire prevention system.

Chapter 2: Methodology

2.1 Data Collection Methods

In this project, data will be collected by using two main methods, namely, the real-time sensor-based data collection and secondary research literature.

An autonomous robotic platform is equipped with many IoT sensors used to collect primary data. These are flaming sensor, MQ-9 gas sensor, DHT11 temperature and humidity sensor, rain sensor and GPS module. This environmental data is continuously read and sent to Wi-Fi by ESP32 which then goes to a cloud server where it is stored and analyzed.

Training and validation of the machine learning fire risk prediction model occur during the system testing phase where simulated environment dataset is created. Patterns of sensor readings are compared to historical sensor readings to determine patterns which belong to various severity groups. The secondary data is obtained through research articles, IEEE journal articles, Elsevier articles, online technical documents pertaining to the field of wildfire monitoring and IoT systems.

2.2 Software Process Model

This project has taken the Agile development approach. The system is built in iterative and incremental sprints that include requirement gathering, design, implementation, testing and evaluation. The Agile method permits the team to keep refining and enhancing the system components based on the testing results and the response of the stakeholders during the development cycle.

Agile has an iterative quality that is especially adaptive to this project since it can accommodate the complexity of hardware, embedded systems, cloud services, machine learning, and mobile development integration. A sprint is dedicated to a particular subsystem, allowing systematic progress and leaving the possibility to include advancements.

2.3 Software Development Tools

The system uses the following tools and technologies in both the hardware and software parts of the system:

Hardware Components:

- ESP32 Wi-Fi Enabled Microcontroller Core Processing and Communication Unit.
- MQ-9 Gas Sensor Carbon monoxide and combustible gas sensors.
- This product is a temperature and humidity sensor called DHT11.
- Flame Sensor -IR flame detector.
- Rain Sensor Precipitation detection.
- GPS Module- Real-time geolocation.
- Servo Motor -control movement of a robot.
- Solar Panel and Rechargeable Battery: Solar power and battery that is environmentally friendly.
- Robot Chassis- Autonomous Mobility Platform.

Software Tools:

- Arduino IDE – ESP32 programming
- Visual Studio Code Web application development.
- React.js – Frontend dashboard
- Firebase / Spring Boot Backend services.
- Python- Machine learning model.
- React Native Mobile application.
- GitHub – Version control.
- Google Maps API - Location visualization.

2.4 Testing Strategies

The strategy of testing is to be considered completely on all system components in order to provide reliability and accuracy:

- Unit Testing: The modules of sensors and software are tested separately to ensure that they can work properly in a controlled setting.
- Integration Testing: The connection between sensors, ESP32 microcontroller, GPS module and cloud services are tested in order to make sure that everything works smoothly.
- System Testing: The system testing is conducted as a whole to ensure the functionality of the end-to-end systems in terms of severity classification logic, delivery of alerts and presentation of dashboard data.
- User Acceptance Testing (UAT): The sample users such as forest officers will test the web dashboard and mobile alert systems to determine how usable and effectively.
- Environmental Simulation Testing: There are diverse fire risk conditions that are simulated by manipulating sensor input to confirm the existence of Low, Medium, and High severity conditions.

2.5 Implementation Plan

The project is done in the following stages in a sequential manner;

1. Literature research and requirement analysis.
2. Technology stack choice and system architecture generation.
3. Purchasing, assembly as well as sensor integration.
4. Embedded programming and sensor data transmission using ES32.
5. Development of cloud backend and database set-up.
6. Implementation of Web dashboard and UI development.
7. Development of mobile applications and integration of alerts.
8. The logic code of Severity Level Classification.
9. Development and integration of machine learning model.
10. Both extensive testing and testing.
11. Proposal to obtain final documentations and submit the project.

2.6 Chapter Summery

The overall approach that was followed in developing the Smart Forest Guardian system was fully described in this chapter. Data collection strategy, Agile process model, hardware and software solutions, multi-layered testing strategy, and organized implementation plan are all strategies that play a role in ensuring systematic, high-quality development and quality system performance.

Chapter 3: Analysis

3.1 Problem and Solution Analysis

The conventional fire detection methods are sluggish and costly and lack intelligent foreboding. The proposed system enhances this with the help of autonomous robots, sensor fusion, machine learning, GPS tracking, and cloud monitoring that help in the accurate detection and prediction of fires.

3.2 UML Diagram

Use Case Diagram

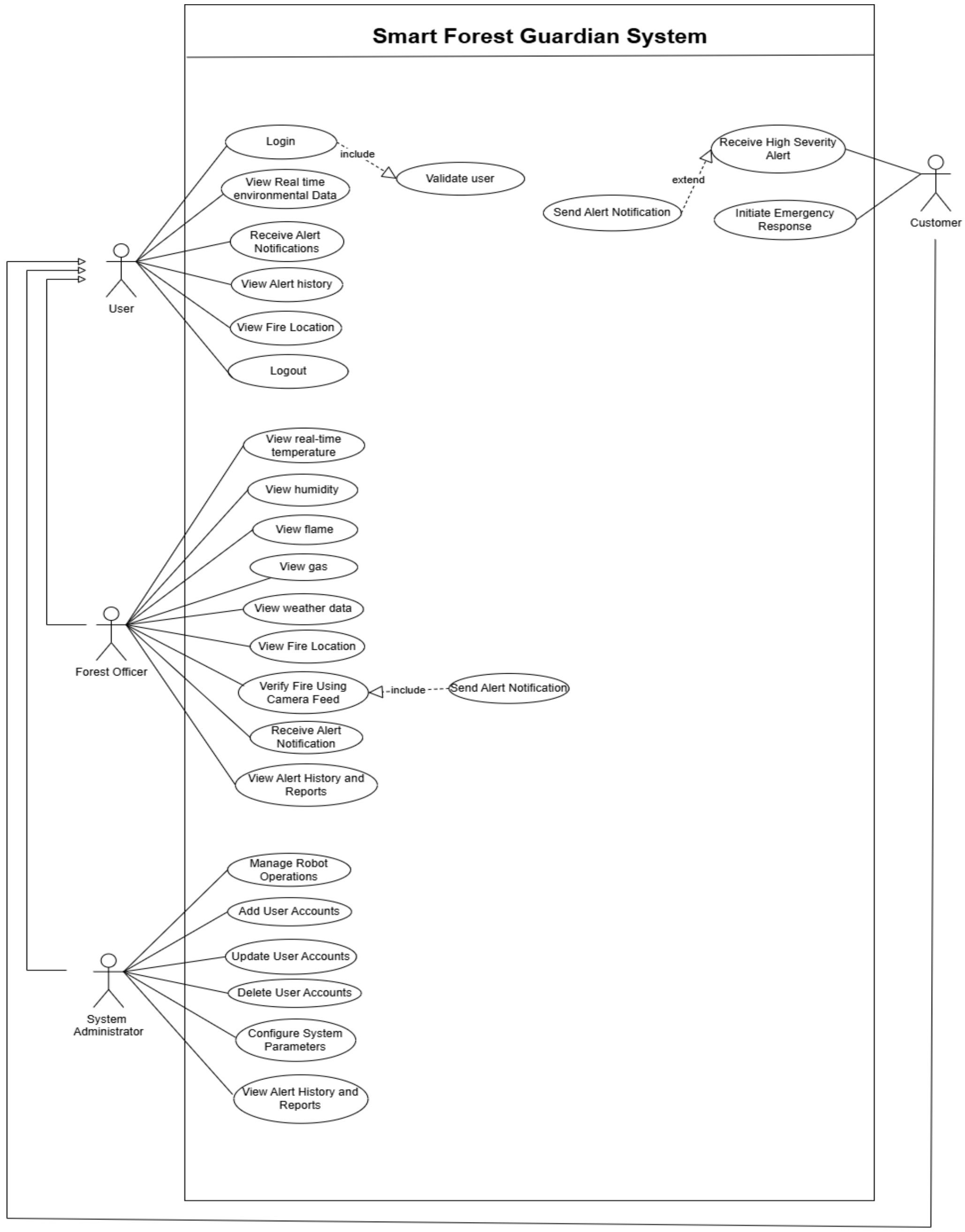

Figure 1

System Diagram

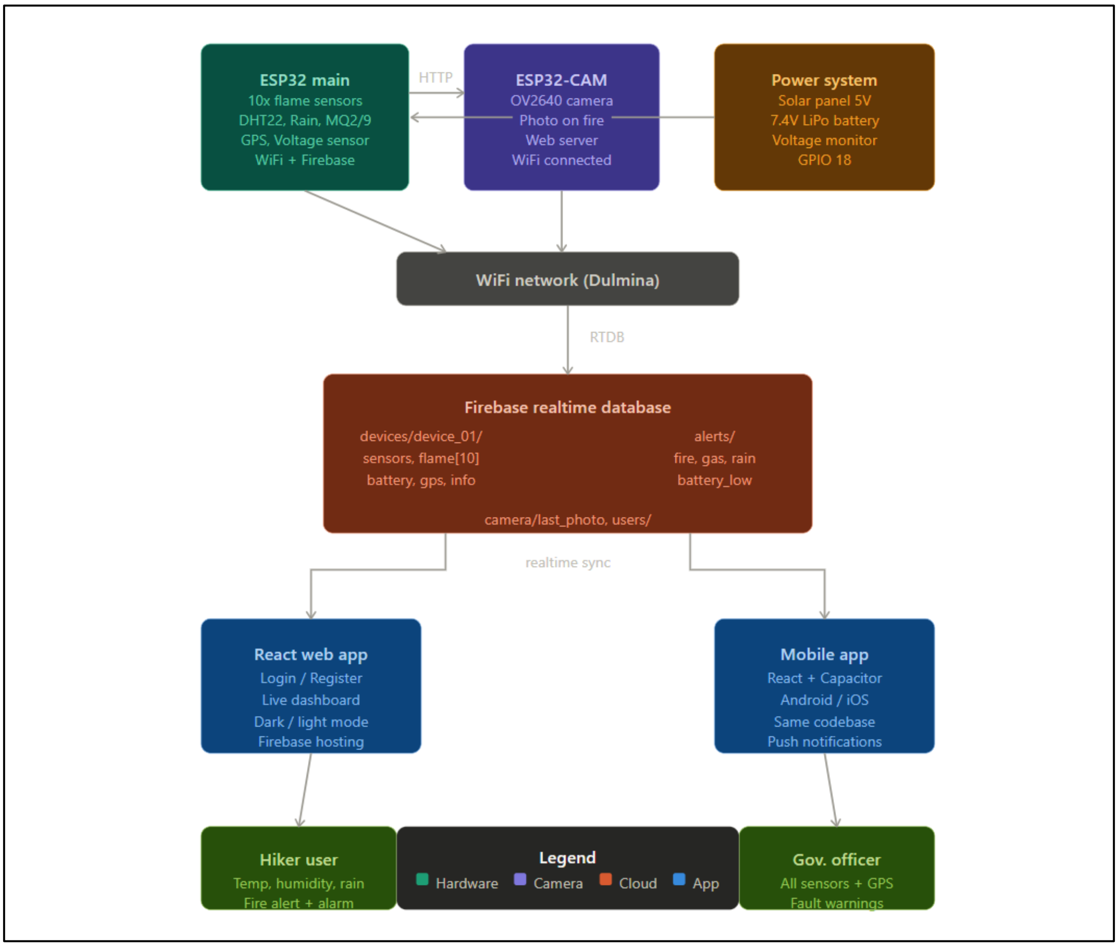

Figure 2

3.3 Data Management of the Proposed System

System contains records in cloud database that include:

- Sensor readings
- GPS coordinates
- Severity levels
- Alerts history
- User data

Frequent backup of data makes data reliable.

3.4 Chapter Summery

This chapter gave a detailed discussion of the problem area, the suggested solution of Smart Forest Guardian, UML system modeling in the form of use case diagrams, and the overall data management approach. The discussion proves the technical feasibility and operational efficiency of the proposed system in dealing with the research gaps found.

Chapter 4: Solution Design and Development

4.1 User Interfaces

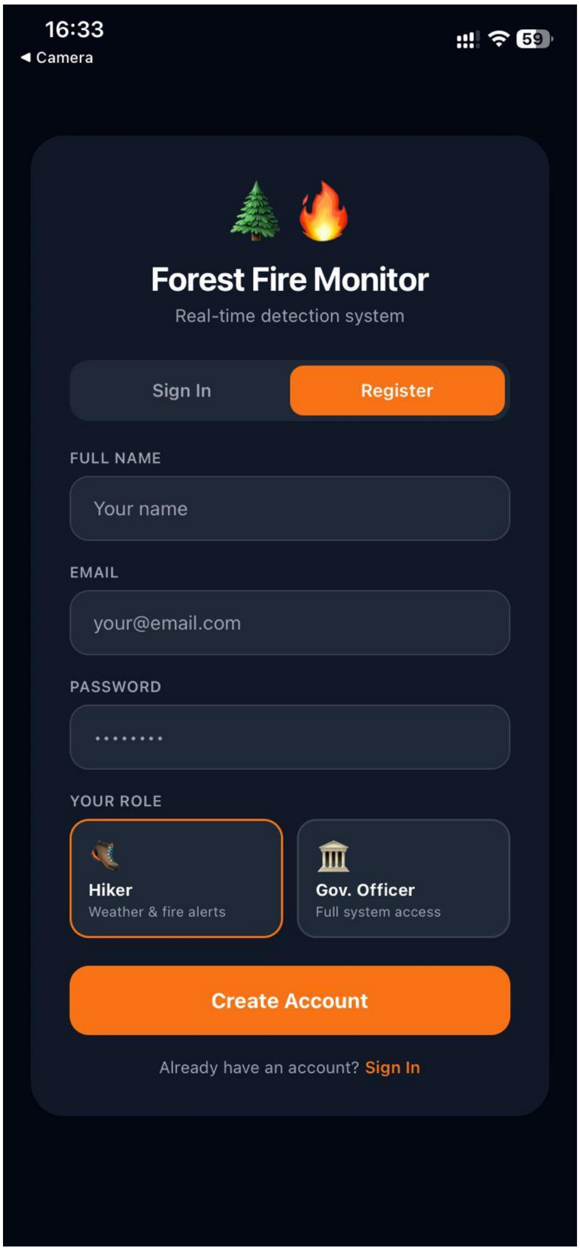

Figure 3

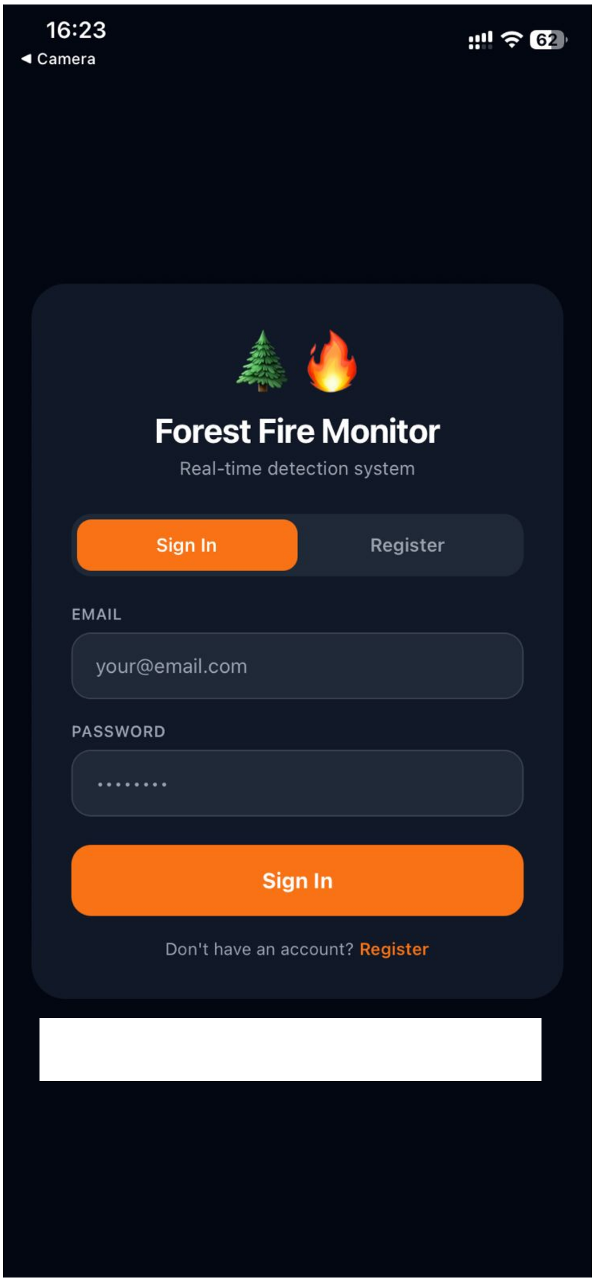
16:23
Camera
62
Forest Fire Monitor
Real-time detection system
Sign In
Register
EMAIL
your@email.com
PASSWORD
Sign In
Don't have an account? Register

Figure 4

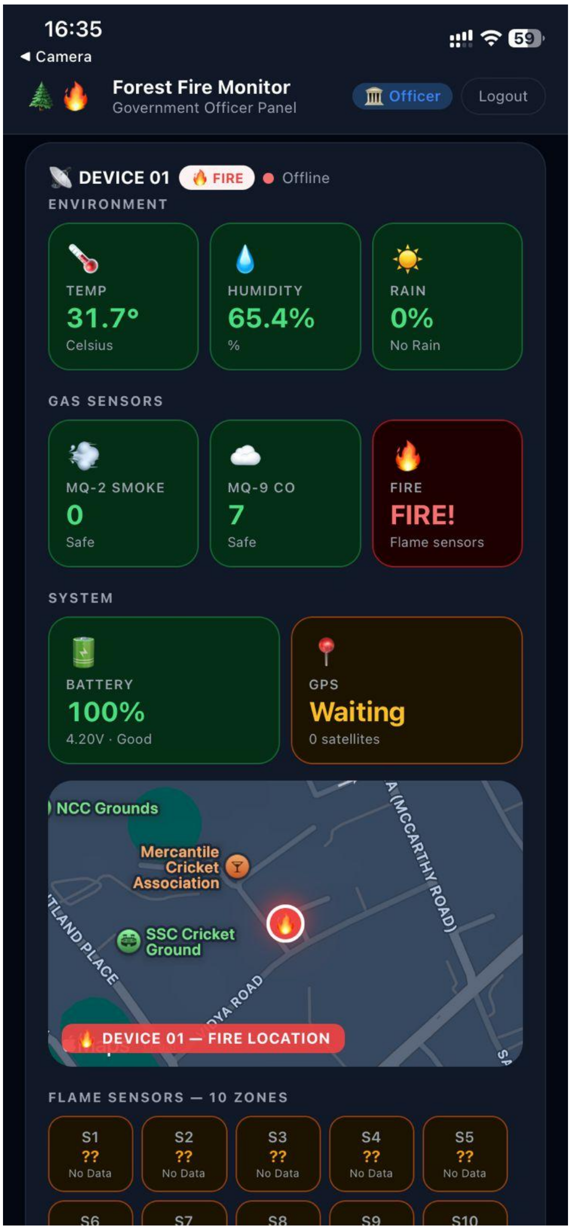
16:35
Camera
Forest Fire Monitor
Government Officer Panel
Officer
Logout
DEVICE 01
FIRE
Offline
ENVIRONMENT
TEMP
31.7°
Celsius
HUMIDITY
65.4%
%
RAIN
0%
No Rain
GAS SENSORS
MQ-2 SMOKE
0
Safe
MQ-9 CO
7
Safe
FIRE
FIRE!
Flame sensors
SYSTEM
BATTERY
100%
4.20V · Good
GPS
Waiting
0 satellites
NCC Grounds
Mercantile Cricket Association
SSC Cricket Ground
DEVICE 01 — FIRE LOCATION
FLAME SENSORS — 10 ZONES
S1
??
No Data
S2
??
No Data
S3
??
No Data
S4
??
No Data
S5
??
No Data
S6
S7
S8
S9
S10

Figure 5

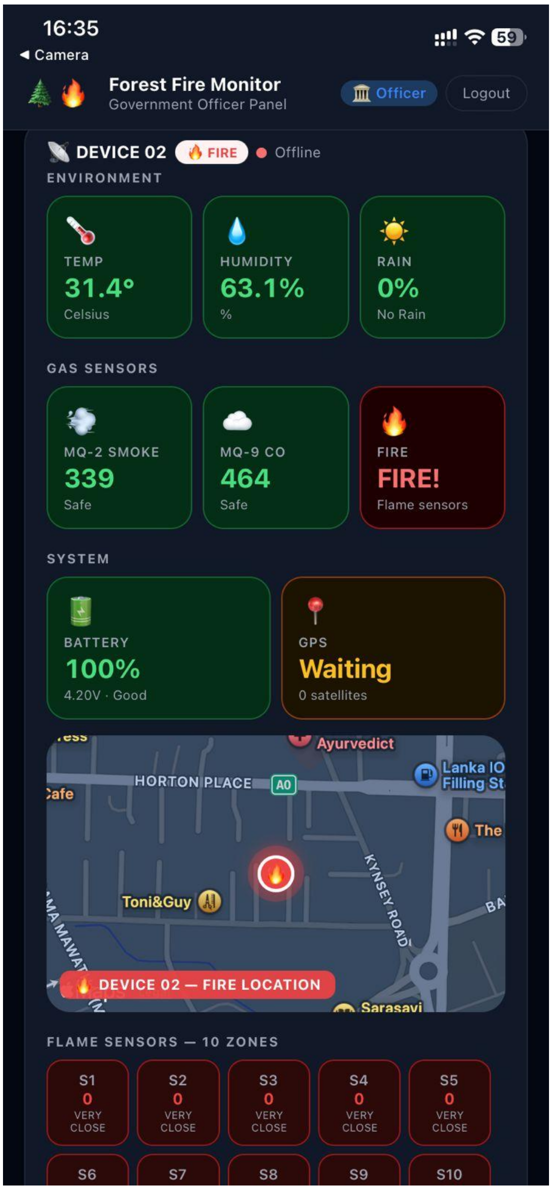
16:35
Camera
Forest Fire Monitor
Government Officer Panel
Officer
Logout
DEVICE 02
FIRE
Offline
ENVIRONMENT
TEMP
31.4°
Celsius
HUMIDITY
63.1%
%
RAIN
0%
No Rain
GAS SENSORS
MQ-2 SMOKE
339
Safe
MQ-9 CO
464
Safe
FIRE
FIRE!
Flame sensors
SYSTEM
BATTERY
100%
4.20V · Good
GPS
Waiting
0 satellites
Ayurvedict
HORTON PLACE
A0
Lanka IO
Filling St
The
Toni&Guy
KYNSEY ROAD
DEVICE 02 — FIRE LOCATION
Sarasavi
FLAME SENSORS — 10 ZONES
S1
0
VERY CLOSE
S2
0
VERY CLOSE
S3
0
VERY CLOSE
S4
0
VERY CLOSE
S5
0
VERY CLOSE
S6
S7
S8
S9
S10

Figure 6

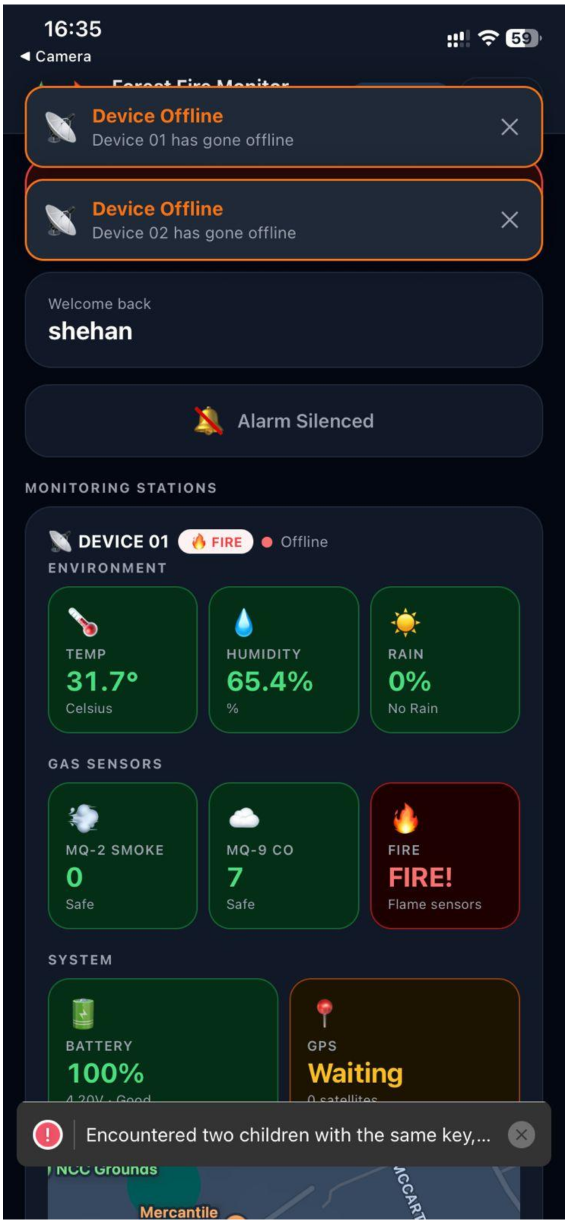
16:35
◀ Camera
Device Offline
Device 01 has gone offline
Device Offline
Device 02 has gone offline
Welcome back
shehan
Alarm Silenced
MONITORING STATIONS
DEVICE 01
FIRE
Offline
ENVIRONMENT
TEMP
31.7°
Celsius
HUMIDITY
65.4%
%
RAIN
0%
No Rain
GAS SENSORS
MQ-2 SMOKE
0
Safe
MQ-9 CO
7
Safe
FIRE
FIRE!
Flame sensors
SYSTEM
BATTERY
100%
GPS
Waiting
Encountered two children with the same key,...
Mercantile

Figure 7

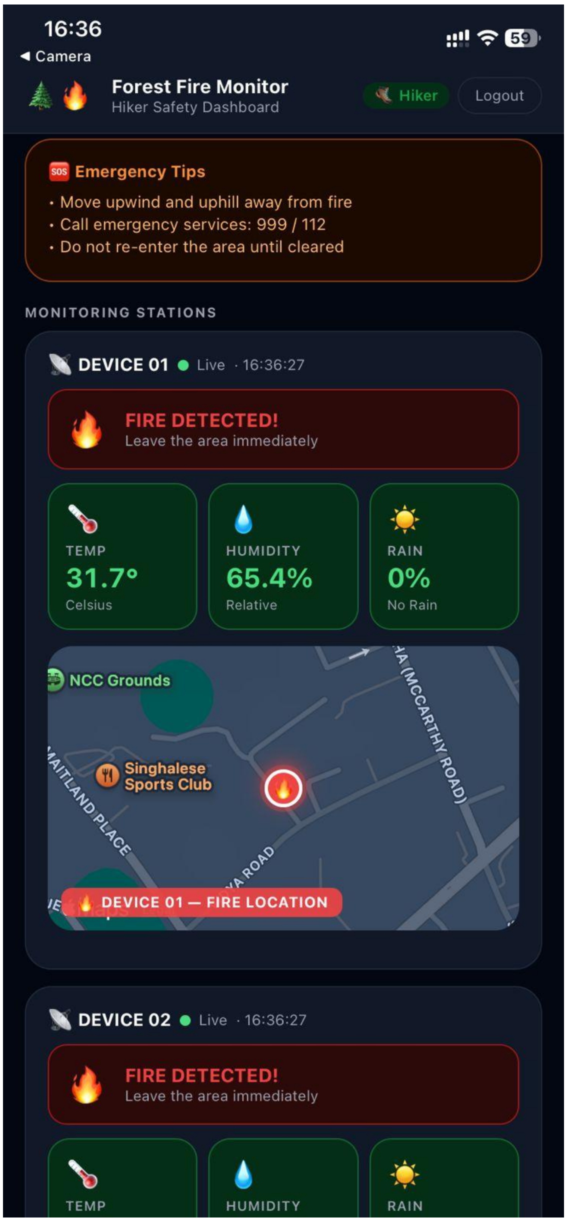
16:36
◀ Camera
Forest Fire Monitor
Hiker Safety Dashboard
Hiker
Logout
Emergency Tips
• Move upwind and uphill away from fire
• Call emergency services: 999 / 112
• Do not re-enter the area until cleared
MONITORING STATIONS
DEVICE 01 Live · 16:36:27
FIRE DETECTED!
Leave the area immediately
TEMP
31.7°
Celsius
HUMIDITY
65.4%
Relative
RAIN
0%
No Rain
NCC Grounds
Singhalese Sports Club
DEVICE 01 — FIRE LOCATION
DEVICE 02 Live · 16:36:27
FIRE DETECTED!
Leave the area immediately
TEMP
HUMIDITY
RAIN

Figure 8

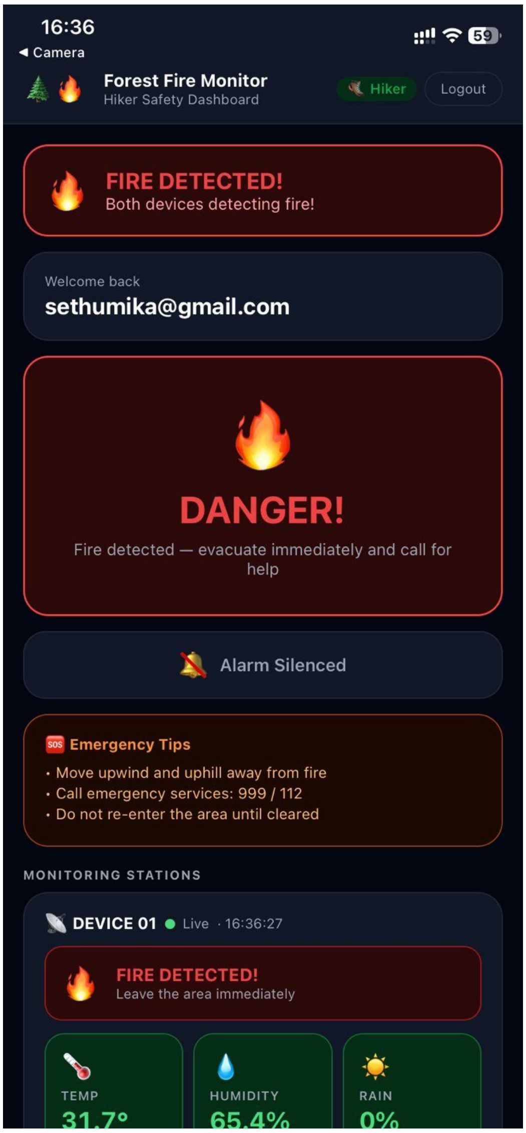
16:36
Camera
Forest Fire Monitor
Hiker Safety Dashboard
Hiker
Logout
FIRE DETECTED!
Both devices detecting fire!
Welcome back
sethumika@gmail.com
DANGER!
Fire detected — evacuate immediately and call for help
Alarm Silenced
Emergency Tips
• Move upwind and uphill away from fire
• Call emergency services: 999 / 112
• Do not re-enter the area until cleared
MONITORING STATIONS
DEVICE 01 Live · 16:36:27
FIRE DETECTED!
Leave the area immediately
TEMP
31.7°
HUMIDITY
65.4%
RAIN
0%

Figure 9

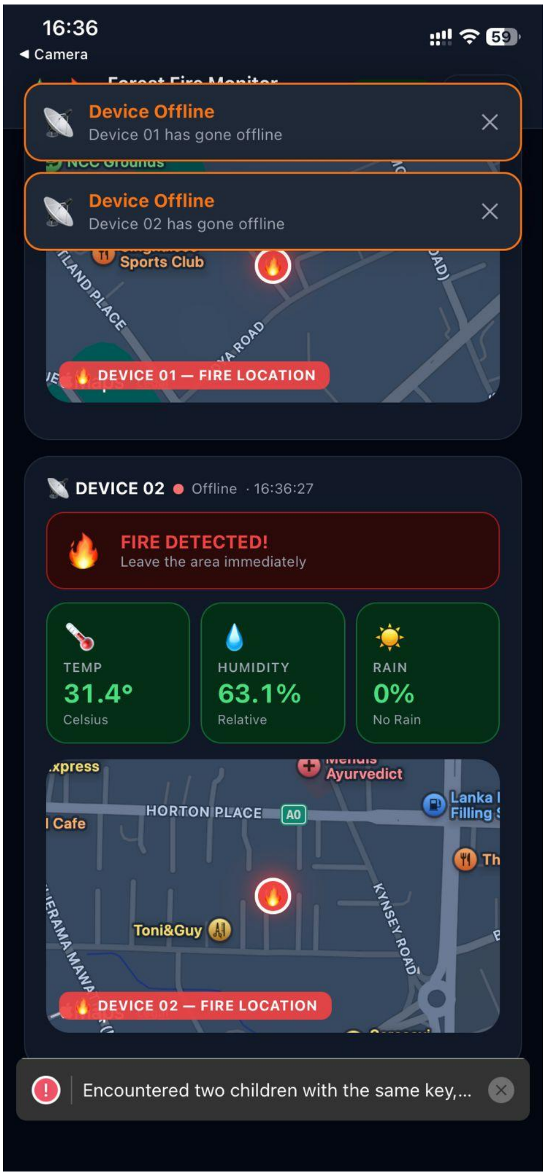
16:36
Camera
Device Offline
Device 01 has gone offline
Device Offline
Device 02 has gone offline
Sports Club
DEVICE 01 — FIRE LOCATION
DEVICE 02
Offline · 16:36:27
FIRE DETECTED!
Leave the area immediately
TEMP
31.4°
Celsius
HUMIDITY
63.1%
Relative
RAIN
0%
No Rain
Ayurvedict
HORTON PLACE
Toni&Guy
KYNSEY ROAD
DEVICE 02 — FIRE LOCATION
Encountered two children with the same key,...

Figure 10

4.2 3D Design of the system

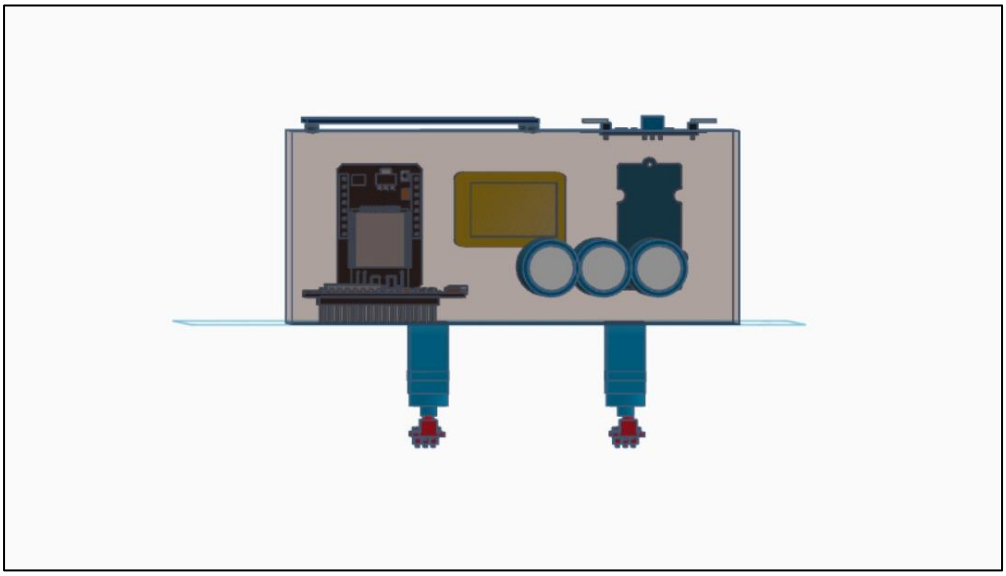

Figure 11

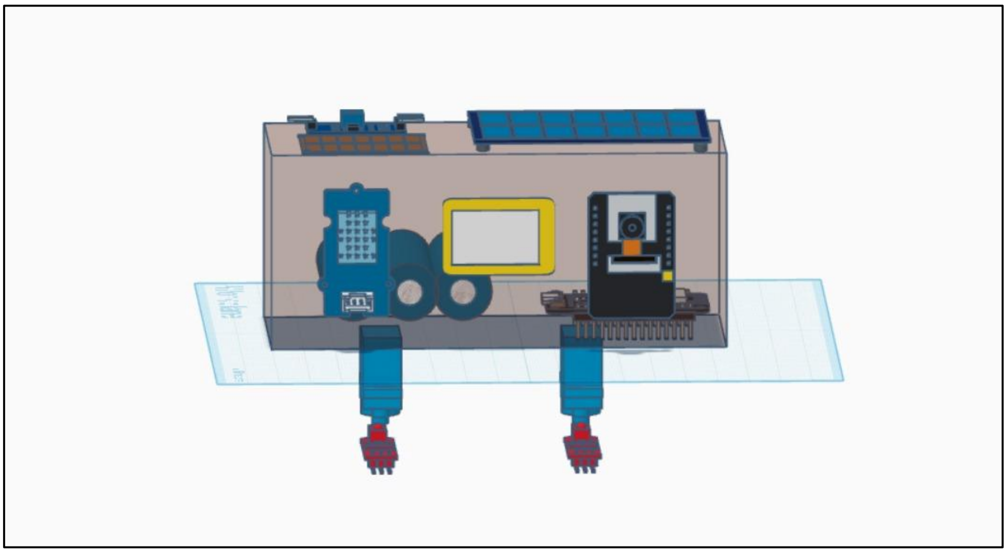

Figure 12

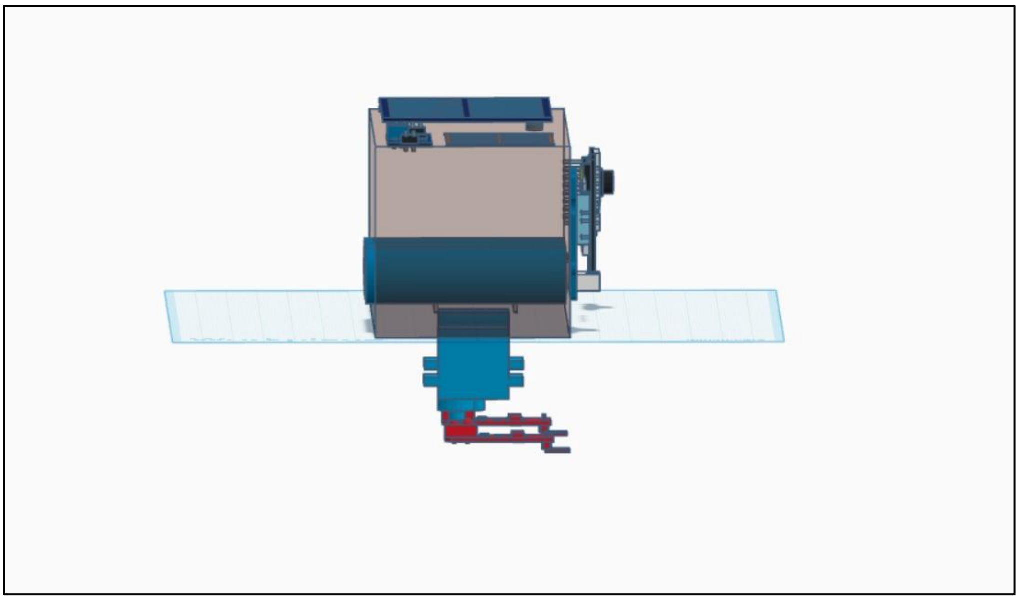

Figure 13

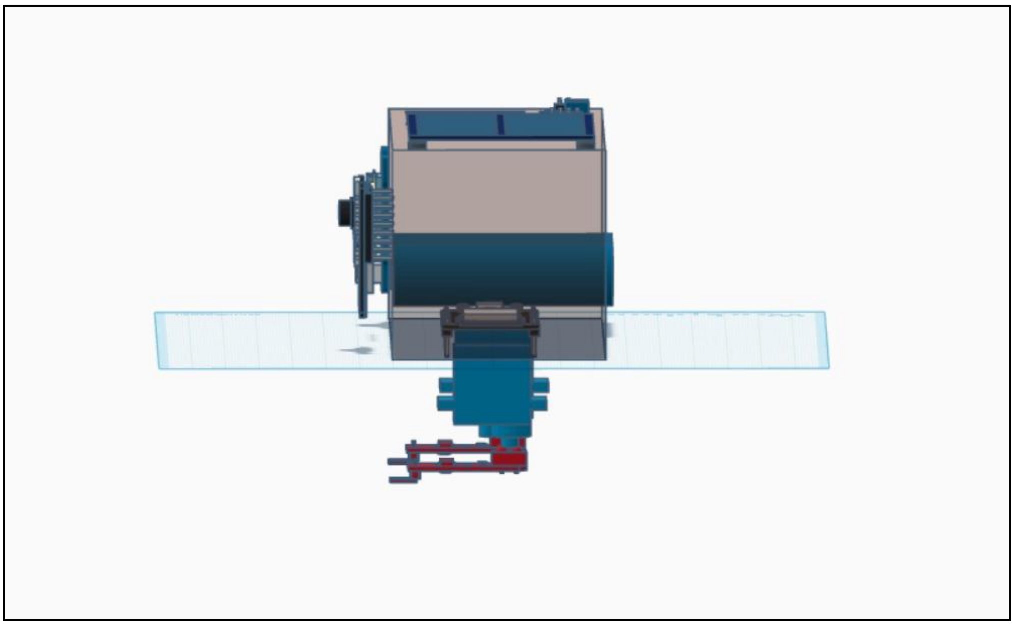

Figure 14

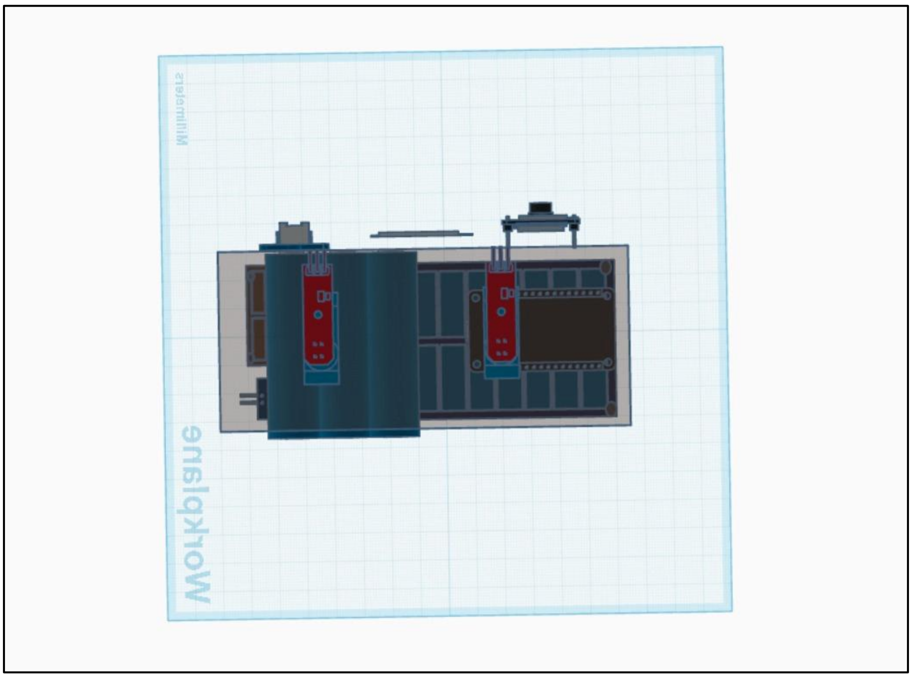

Figure 15

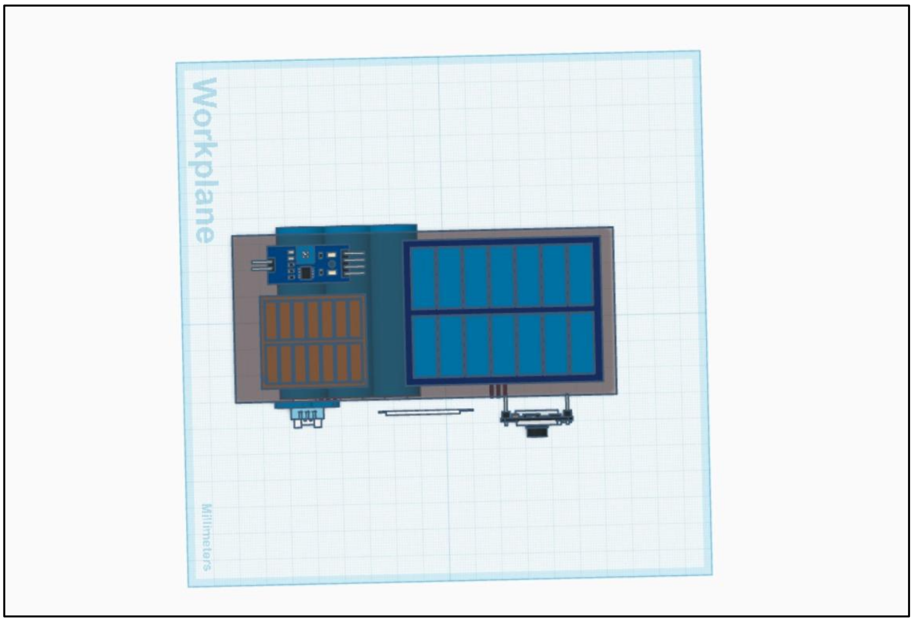

Figure 16

Chapter 5: Cost, Market & Implementation

5.1 Hardware Cost Estimation

Components	Quantity	Unit Price	Total (LKR)
ESP32	1	3500	3500
MQ-9 Gas Sensor	1	1500	1500
Flame Sensor	1	600	600
DHT11 Sensor	1	500	500
Rain sensor	1	700	700
GPS Module	1	3000	3000
Servo Motor	1	1200	1200
Solar Panel + Battery	1	6000	6000
Robot Chassis	1	4000	4000
Jumper Wires & PCB	-	1000	1000

Table 1Cost estimated table

Total cost = 23,000 LKR

5.2 Target Users

- Forest Departments
- Environmental Authorities
- Disaster Management Units

5.3 Market Segments

Government environmental agencies and wildlife monitoring organizations.

5.4 Implementation Strategy

Pilot testing → Improvements → Field deployment → Scaling

Chapter 6: Discussion and Conclusion

6.1 Discussion

The Smart Forest Guardian system is a major step towards solving the long-standing problems that have been related to the detection and control of forest fires. Multi-modal integration of IoT sensors into one autonomous robotic platform offers ground-level knowledge of the environment that cannot be at all offered by satellite and camera-based solutions, especially when such a system is used in highly forested areas and therefore has less aerial visibility.

The implementation of the Agile development system has allowed the team to develop and optimize each aspect of the system gradually so that hardware and software subsystems are well-tested beforehand before joining. The sensor fusion method has demonstrated itself as providing a good level of detection reliability and reduction of the false alarm rate as it has been stated as a key performance requirement during the stakeholder consultations.

The addition of machine learning to predictive fire risk assessment makes the system more than the threshold-based approach to detection and predictive analytics, which will allow prompt instead of reactive emergency actions. The Severity Level Classification System offers a simple and practical intelligence which can greatly enhance the efficiency and effective organization of the emergency response efforts.

The solar-powered design solves one of the most important practical constraints of current forest monitoring systems the impossibility to maintain long-term independent functioning in remote off-grid areas. The Smart Forest Guardian does not only have a technical advantage but also makes it economically viable to use on a large scale and institutional context due to this aspect of sustainability.

6.2 Conclusion

The Smart Forest Guardian project provides a revolutionary solution to the area of wildfire prevention and detection that would bring together autonomous robotics, IoT sensor networks, artificial intelligence, and cloud computing into a single, integrated, and sustainable platform. Unlike the traditional fire detection systems that are only sensitive to fire after it has already begun to establish, the system is geared towards incessant patrols in the forest areas, identify potential risks of fire before it starts spreading and give the relevant authorities accurate and precise information to act promptly and efficiently.

Severity Level Classification System is used to make sure that the ratio of emergency resources to the threat level is proportional and the firefighting and conservation are maximized. Its autonomous design which utilizes solar energy ensures that it has a full coverage of monitoring even in the most remote areas without the need to be maintained or dependent on grid power.

Altogether, it can be concluded that the Smart Forest Guardian is a technically advanced, cost-effective and ecologically friendly solution to one of the most acute ecological problems in our time. Its effective creation and implementation will have a beneficial impact on ensuring the safety of the Sri Lankan Forest ecosystems and can be extended and modified to forest protection activities worldwide.

Chapter 7: References

Kumar, P. (2020) *IoT-based forest fire detection using flame and temperature sensors*. International Journal of Research in Automation.

Liu, F. (2019) *Forest monitoring robot with GPS tracking*. IJRA.

Rao, M. (2022) *Gas sensor applications in wildfire monitoring*. IEEE Sensors Journal.

Patel, H. (2021) *MQ-9 gas sensor applications*. Sensors Review.

Banerjee, S. (2019) *ESP32 in IoT applications*. International Journal of Smart Applications.

Jain, A. and Sharma, K. (2021) *Machine learning approaches for wildfire prediction*. IEEE Access.

Singh, R. (2022) *Severity classification in disaster management systems*. Springer.

Alotaibi, S. (2020) *Cloud-based environmental monitoring using IoT*. Elsevier.

Chapter 8: Appendices

8.1 Questionnaire / Interview Questions

1. What is the current practice of monitoring forest fires in your organization?
2. What are your issues with early fire detection?
3. Is severity-based alert helpful in decision making?
4. What is the value of GPS location during responding to an emergency?
5. Would mobile alert systems enhance quicker response?
6. What were your expectations regarding a forest monitoring system?
7. Do you think about fire monitoring through the use of autonomous robots?

8.2 Reference Documents

- ESP32 Datasheet
- MQ-9 Gas Sensor Documentation
- DHT11 Sensor Manual
- GPS Module Technical Guide
- Arduino IDE Documentation
- Firebase Documentation
- React.js Official Documentation